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Experimental workshops at the Climate HubLAB: a case study on learning spaces for experiential learning through pedagogy and technology

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Introduction: Digital tools for building performance analysis are becoming increasingly essential and widely employed in architectural practices worldwide. However, studies show that while providing useful numerical outputs in architectural design, the extensive use of digital tools in architecture education has the potential to restrict the students' design actions, limit their capacity for innovative thinking, and ultimately lead to design fixations. In contrast, a large body of research gives evidence that many traditional analog practices, such as sketching or physical model-making, are still irreplaceable tools for processing and retaining theory while building tacit knowledge. This paper explores the Climate Hub-LAB at NTNU as a case study for supporting experiential learning practices where students engage in free experimentation and where failure is instrumental in fostering tacit knowledge. The lab merges a traditional design studio with a fully equipped research laboratory with the purpose of supporting hands-on experimentation that combines analog and digital tools to develop more holistic and sustainable design solutions.

Methods: Drawing on Kolb's Experiential Learning Theory (2021) and Schenck and Cruickshank's Co-Constructed Developmental Teaching Theory (2014), the paper examines the correlation between spaces within the lab, the employment of innovative pedagogical approaches, and the combination of analog and digital tools for environmental design. This has been achieved by designing and conducting a series of experimental workshops, under the project DigiHands, over the past few years. These workshops were designed to encourage students to engage actively in their design process and to monitor their learning trajectories and outcomes.

Results: Using Radcliffe's Pedagogy-Space-Technology (PST) Framework for Designing and Evaluating Learning Places (2009), the outcomes from these workshops serve as the basis for relating spaces, activities, and the employment of tools for bioclimatic design practices.

Discussion: Ultimately, the paper reflects on the potential of reimagining architecture studio spaces to enable students to take ownership of their learning and adapt to the evolving architectural profession in the face of rapid digitalization, environmental pressures, and generational shifts.

KEYWORDS
analog tools, pedagogy, digital tools, experiential learning, learning spaces

1 Introduction

This case study examines the space of the Climate HubLAB at NTNU, an experimental learning environment designed to test and analyze innovative pedagogies for deep modes of learning among architecture students. The laboratory was developed as a pedagogical tool for the Climate and Built Form course, where students are exposed to various design problems through a sequence of pedagogical modules that focus on climate-adaptive design. Over the last 2 years, two different experimental workshops have been held at the Climate HubLAB, supported by NTNU through the DigiHands project. One of the purposes of the workshops was to examine the learning potential of the space and its capacity to support experimental pedagogies. The lab itself was envisioned with the Climate and Built Form class's activities and projects in mind. Furthermore, it was designed as a hybrid space, merging elements of a traditional architect's studio with an experimental laboratory, equipped with various analog tools for environmental analysis. By creating this hybrid space, we aimed to encourage students to develop more integrated design processes, in which architectural design of buildings is driven by evidence from analyses or experiments.

The main inspiration for this investigation stemmed from over 10 years of teaching experience in the Climate and Built Form class. These experiences have consistently shown that digital tools, primarily software, often detach students from hands-on activities such as sketching or building physical models, leading them to trust the software more than their own instincts (Finocchiaro, 2018). Oftentimes, this detachment from the physical experience of designing leads to limited creativity and Design Fixations (Jansson and Smith, 1991). Students attending the course are typically novice users of the software used in the classes and struggle to build complex geometries and models, which further discourages them from exploring more creative ideas. Through the intentional integration of analog simulation technology within the Climate HubLAB, we aim to provide Analog Breakouts (Trempe, 2019) as an alternative to the established software used in the Climate and Built Form course, as part of the architect's education. These analog alternatives create space for hands-on activities, physical model making, and spontaneous discoveries, encouraging the development of tacit knowledge. They also facilitate integrating numerical concerns into more creative design processes. However, the intention was not to replace the software's capabilities for environmental analyses, but rather to combine the use of these digital tools with hands-on activities that can more easily foster transformative learning practices among students.

What makes the Climate HubLAB different from a traditional studio is that it has evolved into a pedagogical experiment in itself—combining analog and digital simulation tools for bioclimatic design within a flexible, hybrid learning space that actively supports reflective practice (Schön, 1992), curiosity-driven experimentation, and the co-construction of knowledge. In the intensive experimental workshops, we employed a carefully designed pedagogy and continuously improved our interventions through Action research. Aiming to foster behavioral and cognitive shifts among participating students, the goal was to enhance their engagement with the design and learning processes while also supporting their autonomy as learners. The case study situates the Climate HubLAB within contemporary theories of experiential learning (Kolb, 2021) and constructivist learning (Schenck and Cruickshank, 2014) and evaluates its efficacy using Radcliffe's Pedagogy–Space–Technology (PST) framework (Radcliffe, 2009) (Figure 1). As described in Figure 1, the (PST) framework consists of three equally important, deeply intertwined elements, where each one of them needs and supports the others. To directly translate this framework to our study: "T" stands for technological equipment, in our case, simulation tools, which were placed in the studio. "S," a learning space, aiming to enhance the pedagogical potential of the studio itself, through Pedagogical interventions "P," such as experimental teaching and learning activities.

Intensive workshops have been arranged in Autumn 2023 and Autumn 2024 with the ultimate purpose of testing alternative ways to combine the use of digital and analog tools in climate-adaptive design processes in a way to support learning of principles for climate adaptation while fostering students' creativity and imagination. The critical question was not only how much technical concern should be embedded in a creative form-finding process, but also when and how such concerns should be introduced in a way that still fosters students' imagination and creativity. Both workshops were structured as a sequence of steps, in which students were asked to solve specific climate adaptation problems using both digital and analog tools. A critical difference between the two, as better described later, was that some of the phases were swapped, namely the intuitive and the analytic phases. Students in 2023 were first asked to define a concept for their project using an intuitive approach, then to expose it to a set of analytical experiments to refine form and construction. On the other hand, in 2024, students were first asked to go through a set of experiments to familiarize themselves with how to correctly handle quantitative parameters in their specific climatic conditions, and then build a concept intuitively by playing with images generated during the analytic phase.

This paper is structured into six chapters, beginning with a general introduction and background. Radcliffe's Pedagogy-Space-Technology (PST) Framework for Designing and Evaluating Learning Places (Radcliffe, 2009) is used to identify discrepancies between conception (chapter four) and implementation (chapter five) of the Climate HubLAB concept. Analyses relied on the results of a set of experimental design workshops. In the methodology chapter (chapter three),

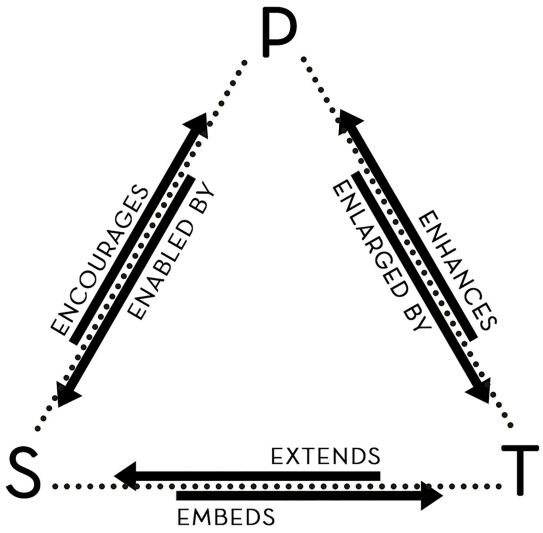


FIGURE 1
Adapted from Radcliffe's PST framework (Radcliffe, 2009). Source: Assembled by the authors.

participants taking part in the workshops are statistically stratified by country of previous education, educational background, and level of knowledge regarding the theory and the tools part of the course, before the criteria for data collection and interpretation are presented. The comparison between conception and implementation serves as the basis for the results described in chapter six, which are then discussed in chapter seven to draw conclusions in chapter eight.

2 Background

The Climate and Built Form course represents the core activity of the first semester of the MSc program in Sustainable Architecture at NTNU. Since its conception in 2011, the course has been structured as a sequence of pedagogical modules, in which students are introduced and exposed to various problems related to climate-adaptive design. In each module, a theoretical part introduces laboratory activities that train students in the use of digital simulation tools. The software implemented in the course enables the execution of complex computational analyses, aiming to facilitate the integration of numerical parameters into architectural design processes. Teaching experiences, however, show that while digital tools provide valid support for embedding technical concerns in creative design processes, they also risk reducing architectural design to a limited number of parameters, failing to reflect its inherent complexity.

Several research works (Ibrahim and Pour Rahimian, 2010; Trnova, 2015; Caper, 1996; Shieh and Chang, 2014) demonstrate that hands-on activities, such as sketching and crafting models, remain an effective way for students to understand new conceptual frameworks and develop the ability to apply them in practice in creative and innovative ways. Sketches and physical models are powerful tools for integrated design processes; however, concepts developed through tacit knowledge can be more challenging to verify with digital tools, as they often involve non-standard solutions. Students, therefore, choose either to follow a purely qualitative approach using analogue tools or to abandon courageous ideas in favor of more convincing quantitative analyses in the digital environment. This tendency limits the full achievement of the course learning outcomes, where skills in digital computational tools are combined with a holistic understanding of the project requirements.

The question, therefore, is whether it is possible to develop a series of teaching and learning activities that actively involve students in hands-on experiences and utilize the computational capabilities of digital tools to enhance their learning potential. The Climate HubLAB was initiated in 2016, thanks to the support of the AVIT program at NTNU, and is designed to offer students an integrated studio environment equipped with various machines and tools for environmental analyses.

In 2020, NTNU supported the development of the DigiHands project as an opportunity to systematically explore the pedagogical potential of the Lab in aiding integrated design processes. The ultimate scope of the DigiHands project was to test, through a set of intensive workshops, the learning environment inside the Climate HubLAB and evaluate its potential to support experiential learning practices and modes of deep learning. The workshops conducted inside the project emphasized the importance of "learning by doing" as a central pedagogical principle. Free experimentation—whether through structured teaching and learning activities or spontaneous play—was not viewed as peripheral but as a legitimate and essential mode of learning. Students were encouraged to view curiosity as a valid starting point for design inquiries and to treat their interests and intuitions as worthy of pursuit. This intentional blurring of formal and informal modes of experimentation allowed for moments of discovery to emerge from unstructured exploration, rather than solely from facilitator-led activities.

3 Methodology

In this paper, we employ the Pedagogy-Space-Technology (PST) Framework for Designing and Evaluating Learning Places, developed by Radcliffe (2009), to evaluate the pedagogical potential of the Climate Hub lab as a learning space through investigating and analyzing the outcomes of the DigiHands project. The original Action Research Methodology employed in the development of the workshops within the DigiHands project uses a mixed-methods approach, drawing on both qualitative and quantitative data.

In Figure 2, the connection between the workshops' methodology and the methodology used to evaluate the learning space in which those workshops took place can be traced. The red circle with the number "4" in it indicates the framework for this paper.

There are several reasons why we have selected Radcliff's PST Framework for this paper. In his paper "A Pedagogy-Space-Technology (PST) Framework for Designing and Evaluating Learning Places," Radcliffe (2009) discusses multiple initiatives and ongoing projects throughout Europe, the United States, Canada, and Australia aiming to develop innovative learning spaces as a response to the generational shifts resulting in changing social patterns, rapid digitalization, and the emergence of new technology, focusing on pedagogical changes towards more learner-centered teaching strategies. In this work, Radcliffe (2009) discusses the differences between various projects and initiatives, emphasizing the different combinations of factors considered. He argues that rarely are all three factors—pedagogy, space, and technology—given equal attention and treated equally important. His paper concludes with a comprehensive framework for evaluating learning spaces that considers all three aspects and seeks to integrate the multitude of recommendations, guides, and design principles proposed by various authors (Radcliffe, 2009).

In 2023, using the PST model in a specific case study, De Caro-Barek et al. (2023) discussed at length the role of space in the learner's experience. Considerations in their case study build on research conducted by Boys (2011) and Radcliffe et al. (2008), according to which research on places of learning too often focuses entirely on the space as a determinant of learning activities at its interior, leaving little agency to the pedagogy, the educator, the learner's ownership of their learning process, and overall learning experience. De Caro-Barek et al. (2023) stress the importance of learning environments, both physical and digital, from a phenomenological perspective, drawing on Merleau-Ponty's Phenomenology of Perception, in which the agency of users and "the life unfolding in the learning environment" are central. Furthermore, they discuss the need for an in-depth conversation on pedagogy in relation to the future of university education and the conception of future learning spaces, arguing for a "Pedagogy-first" approach that builds on Radcliffe's Pedagogy-Space-Technology framework (Radcliffe, 2009) and extends it (De Caro-Barek et al., 2023).

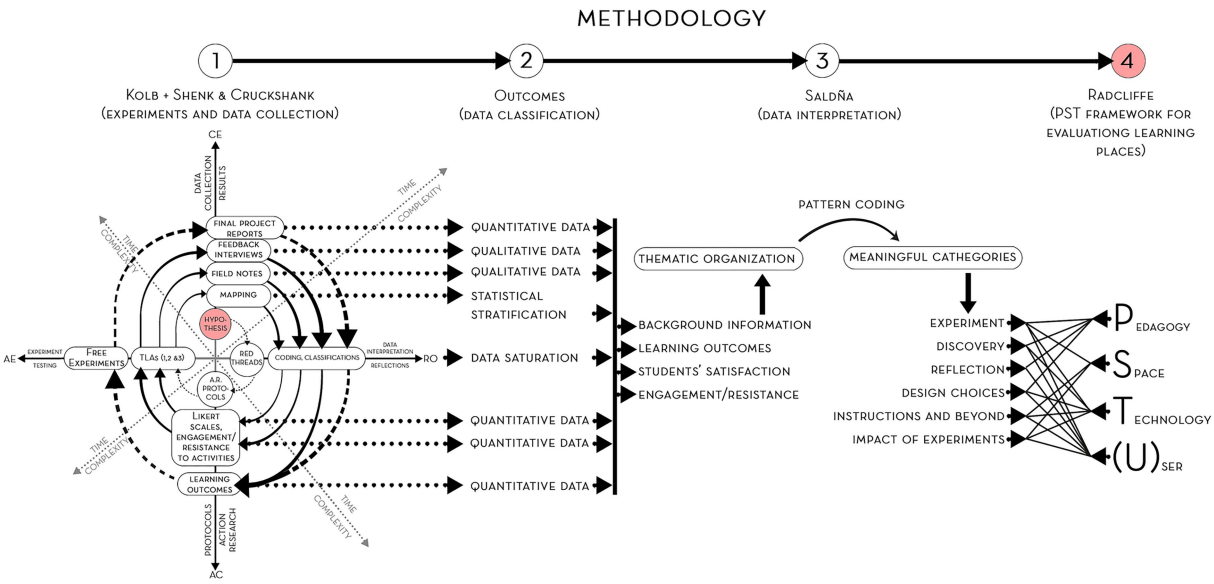


FIGURE 2
Methodology of this paper, building on the methodology employed in the DigiHands project. Source: Assembled by the authors.

In a book titled *"Designing Hybrid Learning Environments and Processes,"* Manciaracina (2022) assumed a critical position towards the PST framework for not explicitly including users in the triad. In the same book, Marancina therefore proposes a four-element model—called PSTU (Figure 3)—where "the user," now placed at the center, assumes a more prominent position, "establishing new connections and actions through the mediation of design" (Manciaracina, 2022). Manciaracina's PSTU Framework (Figure 3), which links the user's role to the original PST triad, was used to reflect on students' roles as users in the development of the experimental workshops in the Climate HubLAB. The revised model emphasizes not only the effects that Pedagogy, Space, and Technology have on users, but also how users, through their involvement in the process, affect Pedagogy, Space, and Technology. For example, feedback from users could imply a revision of pedagogical approaches implemented in the workshop, suggest new ways in which technical equipment included in the studio can be used in more creative and innovative ways, and finally allow us to see how the learning space can be more carefully designed in a way to let users transform it, enhance the learning potential and take ownership.

In this article, we analyze and reflect on the role of the Climate HubLAB as a tool for supporting learning in environmental design processes. The PST framework was used as an analytical tool to examine the relationships among pedagogy, space, and technology in our research activities, emphasizing their interaction. While acknowledging the model's limitations regarding users and pedagogy, as stressed by the aforementioned research, we still followed the original PST model (2009). Concerns raised by De Caro-Barek et al. (2023) and Manciaracina (2022) were taken into account by putting users and pedagogy at the center of our teaching and learning activities, first, and analyses, later. The PST model, as summarized in Table 1, facilitated the identification of agents responsible for a discrepancy between the concept and the implementation of the original Climate HubLAB idea. Our answers to the PST framework's matrix of questions are based on the results and feedback we collected from students who participated as users in the intensive workshops developed as part of the DigiHands project.

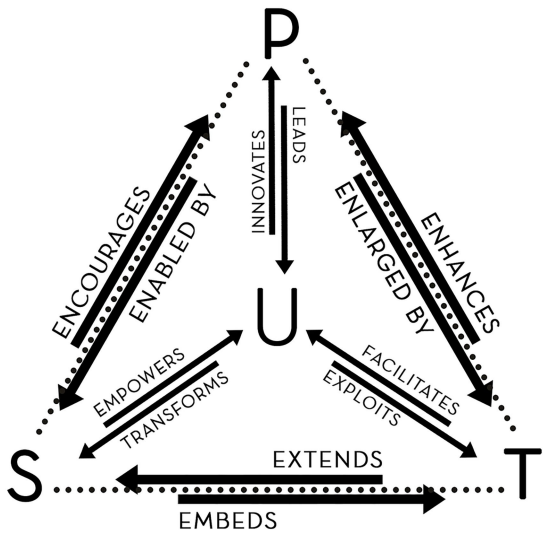


FIGURE 3
Adapted from Manciaracina's PSTU framework (Manciaracina, 2022). Source: Assembled by the authors.

3.1 Participants

Between the two workshops conducted in 2023 and 2024, a total of 50 participants took part in the experimental design workshop. All participants were master's students enrolled in the multidisciplinary MSc program in Sustainable Architecture with diverse cultural and educational backgrounds. About 24% of students were from Norway, while the remaining 76% were international students with proficiency in English, as required by the program. Before the workshop, all students filled out anonymous mapping sheets about their educational background, the country of their prior education, the level of previous knowledge of the digital and analog tools introduced in the workshop, and their prior knowledge of bioclimatic design and environmental

TABLE 1 Adapted from Radcliff's PST design and evaluation criteria (Radcliffe, 2009).

Focus	Concept and design	Implementation and operation
Overall	What is the motivation for the initiative? What is intended? What initiated the project? Who are the proponents and opponents? Who has to be persuaded about the idea? Why? What lessons were learned for the future?	What does success look like? Is the facility considered to be a success? By whom? Why? What is the evidence? Does this relate to the original motivation or intent? What lessons were learned for the future?
Pedagogy	What type(s) of learning and teaching are we trying to foster? Why? Why is this likely to make a difference to learning? What is the theory & evidence? What plans will be made to modify programs or courses to take advantage of the new facilities? What education or training for academics and other staff is built into the plan?	What type(s) of learning and teaching are observed to take place? What is the evidence? What evaluation methodology or approach was used, and what methods were used to gather and analyze data? Who was included in the data gathering and analysis? Students? Faculty? Staff? Administrator? Senior Leadership? Facilities managers and technology staff?
Space (including environs; furniture and fittings)	What aspects of the design of the space and provisioning of furniture and fittings will foster these modes of learning (and teaching)? How? Who is involved in developing the design brief? Why? Which existing facilities will be considered in developing concepts? Can we prototype ideas? Who is involved in the assessment of concepts and detailed design? Why? What are their primary issues and concerns?	Which aspects of the space design and equipment worked and which did not? Why? What were the unexpected (unintended) uses of the space and facilities that aided learning or facilitated teaching? Do these present ideas for future projects? How was the effectiveness of the use of space to aid learning and teaching measured? What were the different metrics used? Where there synergies between this and other spaces that enhanced learning?
Technology (lab and specialist equipment)	What technology will be deployed to complement the space design in fostering the desired learning and teaching patterns? How? In establishing the brief and developing concepts and detailed designs, what is the relationship between the design of the space and the selection and integration of technology? What pedagogical improvements are suggested by the technology?	What technologies were most effective at enhancing learning and teaching? Why? What were the unexpected (unintended) impacts (positive and negative) of the technology on learning and teaching? How did technology enhance the continuum of learning and teaching across the campus and beyond?

performance. The results of data collection are synthesized in Table 2. Ethical approval for collecting this data was not required, as no names, ages, or other distinguishing information were collected. Students also completed feedback reports using Likert scales and charts to map their satisfaction with the various teaching and learning activities from the workshops and the usefulness of the tools embedded in those activities.

3.1.1 Sample size and distribution

The number of participants in the workshop is constrained by the number of people enrolled in the Master's program in Sustainable Architecture each year. Since the study is ongoing and more workshops are planned for the future with slight variations to test nuanced situations, we cannot conclude that full data saturation has been reached. However, Code saturation in the second round of workshops was not expanded (indicating it may have been accomplished). In contrast, Meaning Saturation has not been achieved (Hennink et al., 2017; Hennink and Kaiser, 2022), and we expect to hear and see more from future workshops that will add valuable insights, helping us better understand students' learning experiences.

Several review and discussion papers (Miller, 2012; Passerini, 2012), as well as Warren's (2002) *Handbook of Interview Research: Context and Method*, suggest that to reach data saturation, the number of interviews required depends on a variety of factors such as "scope, target audience, researchers, resourcing and the research audience" (Bonde, 2013). However, the guidelines those studies provide for

TABLE 2 Sample size and distribution of the participants.

Workshops	1st - 2023	2nd - 2024
Number of participants	20	30
The number of countries the participants were from (outside Norway)	10 Belgium, Brazil, China, Denmark, Germany, Iran, Italy, Netherlands, Poland, Spain	15 Austria, Belgium, The Czech Republic, Denmark, France, Germany, Italy, Kurdistan, Latvia, Palestine, Philippines, Switzerland, Türkiye, Ukraine, The United States
From Norway	35%	16%
Background	Arch. - 70% Eng. - 20% Arch./Eng. -10%	Arch. - 67% Eng. - 13% Arch./Eng. - 20%
Prior knowledge of the digital tools used in the course	No knowledge - 60% Some knowledge - 40%	No knowledge - 73% Some knowledge - 27%
Prior knowledge of the theory introduced in the course	Good - 5% Basic - 55% No knowledge - 40%	Good - 20% Basic - 57% No knowledge - 23%
Prior use of the analog tools used in the course	N/A	1 person (wind tunnel)

sample size variations required to reach data saturation range from 1 to 100. Considering the mixed methods nature of this study, focusing on the learning experiences of master's students (qualitative data), and the quality of the learning outcomes they reach and produce throughout the completion of their semester projects (quantitative data), we considered the number of 50 students as sufficient to draw first conclusions of our study and potentially have reached code saturation.

3.2 Data collection

A large volume of data was collected and recorded during and after the experimental workshop within the DigiHands project. However, this paper aims to discuss the learning potential of the space and its capacity to support experimental pedagogies through technology. For this reason, only data relevant to answering this specific question is included in the data analysis and interpretation for this paper.

The DigiHands project employed a mixed-methods approach to developing the workshops, and data collection protocols were established at the outset. The data collection protocols enabled us to impose constraints on the collected material, thereby placing the data in context. We referred to these constraints as 'red threads,' which were strategically integrated into the data collection protocols and later served as a 'start list' (Saldana, 2013) for the first cycle of the data coding using a Provisional Coding method.

Both qualitative and quantitative data were collected and interpreted. The data included in this paper consists of:

- Mapping sheets for the students' backgrounds for statistical reasons (Quantitative data)
- Likert scale charts allowing the students to rate each activity and the tools they employed (Quantitative data)
- Field notes from semi-structured round table discussions and interviews with the students, notes from each presentation, concluding every workshop cycle, and notes recording any changes to the studio (Qualitative data)
- Sheets for open feedback from the students regarding the activities and the tools (Qualitative data)
- Project reports delivered at the end of the workshops were evaluated for the quality of the experiments conducted, the results yielded from them, and how they affected the students' project design development (Quantitative and Qualitative data)

Ethical approval for collecting this data was not required, as no names, ages, or other distinguishing information were collected.

3.3 Data interpretation

Although the data collected from both workshops were comparable, the coding methods varied slightly due to differences in the workshop formats. Nevertheless, all data were manually coded using a range of methods appropriate to the Action Research framework and the nature of the data collected.

For the first workshop, coding was conducted in three distinct cycles. The data were thematically organized (Saldana, 2013) and reflected upon using analytical memos, which were then visually represented in matrices (Table 3). A separate matrix was created for each workshop cycle to facilitate comparison and contrast of the outcomes derived from using different tools. These matrices were also aligned with the steps of the experiential learning cycle (Figure 4, point 1) to ensure consistency. By structuring the data in this way, we were able to examine the extent to which each student group engaged with the learning cycles—whether they conducted experiments, interpreted the results, reflected on them, and whether these reflections informed their design decisions.

The first coding cycle employed Provisional Coding (Saldana, 2013), based on a predefined "Start List" of "Red Threads" (ibid.), which we aimed to trace throughout the workshop. The preliminary categories for the Provisional coding were "Background information" used for statistical reference, "Learning Outcomes," focusing on whether the workshop activities shaped the design process and helped students enhance their work systematically using evidence, "Students' Satisfaction" with the various teaching and learning activities (rated as helpful, not helpful, neutral, or no response), and "Engagement with" or "Resistance to" the different tools. These last two categories addressed how students used the tools, the ways they applied them, and whether they adhered to the given instructions.

TABLE 3 Example of an evaluation matrix for the workshop activities.

Groups	G1	G2	G3	G4	G5
1. Experiment					
– Unplanned free experimentation (in order to discover unexpected things)					
– Planned experiments (testing of conditions)					
– Planned experiment (testing existing design)					
– Planned experiment (testing of solutions)					
2. Discovery					
– Something surprising/unexpected					
– Something opposite to expected					
– Got confirmation on speculation/theory					
– "Saw the bigger picture"					
– Other					
– Didn't discover anything					
3. Reflections					
– Critical reflection on own design					
– Reflections on discoveries					
– Reflections on possible solutions					
– Reflections on initial objectives					
4. Design choices					
– Due to a discovery					
– Due to an experiment					
– Due to a reflection (compromise)					
– Confirmation of initial design choices					
5. Followed instructions thoroughly					
6. Module influenced the design development					

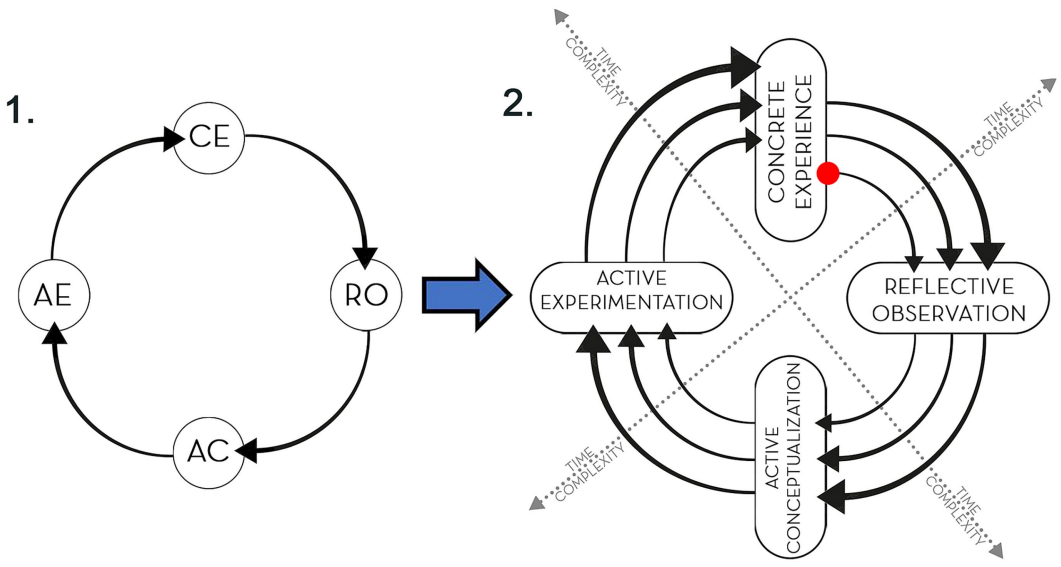


FIGURE 4
1. Adapted from Kolb's learning cycle; 2. Adapted from Schenck and Cruickshank's co-constructed developmental teaching theory. Source: Assembled by the authors.

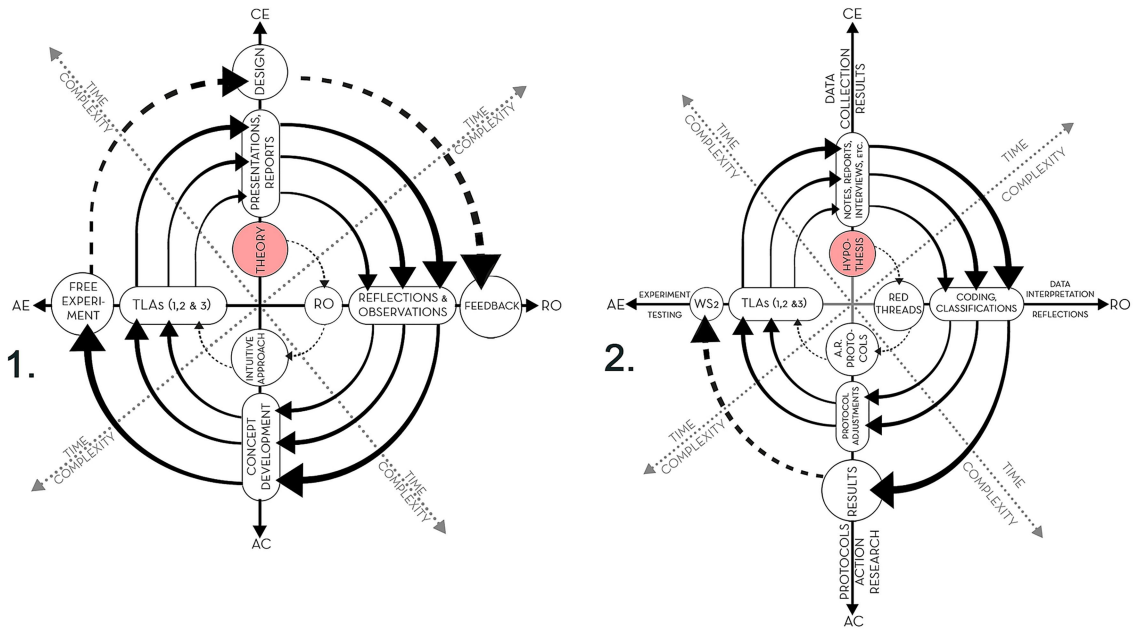


FIGURE 5
1. Workshop activities' scheme; 2. Research design structure scheme. Source: Assembled by the authors.

While some provisional codes persisted throughout the process, others emerged from the empirical data during the first cycle when the Initial Coding approach was repeated. During this second iteration, we incorporated *In vivo* and Process Coding to ensure the findings were closely grounded in the data retrieved.

The final stage involved a second cycle using Pattern Coding, which enabled us to split, splice, and link the data into meaningful categories (Saldana, 2013). These final categories informed the evaluation tables (Table 3), through which we analyzed the depth of student engagement in the learning activities, the degree to which they were immersed in the learning cycles, and whether they successfully entered and re-entered different stages of those cycles.

The data from the second workshop were coded in two cycles, using the matrices from the first workshop. However, those matrices were simplified, and some categories were not taken into consideration, because the second workshop was organized with stricter closed constraints to ensure that all students went through every step of the learning cycles with every tool (Figure 4, point 1). Furthermore, the timeframe was also shorter, leaving little to no space for variations in the activities.

Field notes were taken on various occasions when students made unexpected changes to the studio, documenting their interactions and adaptations as they took ownership of the space. The content of these notes is also included in our answers to the matrix of questions (Table 1) following the PST model (Figure 1).

4 Conception and design

The Climate HubLAB was initiated in 2016 to provide students with an integrated studio environment equipped with various simulation machines and tools for environmental analysis. Since then, the Lab has been continuously developed, expanding the technical equipment included and rearranging the layout to support evolving teaching and learning activities. In this chapter, the pedagogical framework, lab space, and equipment are described as originally conceived. The lab's equipment facilitates a seamless flow of information between digital and analog tools, providing input for more efficient architectural design processes in which concepts are conceived through hands-on experience. The Climate HubLAB served as a platform to facilitate a synthesis of quantitative and qualitative parameters in environmental design processes, enabling us to test specially designed pedagogical approaches that incorporated analog and digital tools for environmental design. The intensive workshops arranged as core activities of the DigiHands project, focusing on pedagogic research in sustainable environmental design, aimed for a behavioral shift in the participating students toward ownership of both their learning and their design processes, enabling them to follow their creative instincts and support their design choices with evidence, to take the initiative to explore, test, and play.

4.1 Pedagogy

4.1.1 Pedagogical interventions and workshop design

A pedagogical strategy was developed and structured as a double-loop pedagogical learning cycle (Figure 5) to explore the potential of balancing the use of analog and digital tools in environmental design. Each workshop was designed to follow an experiential learning cycle, adapted from Kolb's Experiential Learning Theory (Kolb, 2021) (Figure 4, point 1). Schenck and Cruickshank's Co-Constructed Developmental Teaching Theory (Schenck and Cruickshank, 2014) (Figure 4, point 2) was used, moreover, as a framework to enable reflection on both the design outcomes and the learning process itself, by turning the learning cycles into a spiral of iterative cycles (Figure 4).

This reflective model was mirrored by a second spiral loop of experiential learning cycles intended for instructors and researchers. In this

way, both students and teachers would go through a process of experimentation, observation, conceptualization, and active experimentation (Figure 5). Through action research methods, the spiral of experiential learning cycles enabled a continuous and structured adaptation of the teaching practice, based on reflections on real-time observations and the students' feedback. This double-loop pedagogical learning cycle provided structure for the workshops' teaching and learning activities and for our own interventions. Each cycle consisted of four steps: 1. Concrete experience; 2. Reflective Observation; 3. Abstract Conceptualization and 4. Active Experimentation. Both the students and the teachers/researchers went through those loops multiple times. The students' activities in both workshops were designed to follow these steps. The students were free to go back and forth through the cycles and repeat each step as many times as they needed. However, the teachers and researchers mostly followed the steps consecutively (Figure 5, point 2).

To understand the relationship between the two loops, they have to be observed in parallel in the way they are presented in Figure 5. The intersection between the students' activities and the educators' loop begins a little before the start of the workshop, both at the red circle in the centers of the two experiential learning spirals. For the students, the red circle represents the theoretical knowledge they have acquired at the beginning of the course. For the teachers, it represents the study's core hypothesis, which examines the combination of digital and analog tools to enhance students' learning experience. In the next step, "Reflective Observation" (RO) (for the students) and "Red Threads" (for the educators), students had to reflect on the theory and apply the knowledge in practice. In contrast, the educators, had to identify what to look for, and to create a start list with provisional codes. Later, the students were asked to start their design process, in the first workshop with an intuitive approach first, and in the second one with an analytical approach first. In parallel, the educators were tasked with developing the research protocols for data collection and interpretation.

The different arrows in the spiral represent the workshop's timeline. The dotted lines represent related activities outside the workshop timeframe, and the bold black arrows represent activities part of the workshop. For the following steps, you can observe in Figure 4, the "TLA's" (the teaching and learning activities), align for both spirals. The three TLA's steps represent the stage of active experimentation for both students and educators. While the students experimented with the different tools, the teachers and instructors were responsible for applying action research methods and adjusting their instruction and facilitation as needed.

Next, on the axes for "Concrete Experience" (CE), you can observe the step "presentations and reports" for students and "notes, reports and interviews" for educators. These steps are structured so that every completed experiment led the students to a step where they had to present what they had accomplished through different means. Those means could be more casual roundtable discussions, small presentations in front of the class, or formal presentations and discussions. For educators, those steps represented an opportunity to collect both qualitative and quantitative data through notes, protocol reports, or interviews. The cycles repeated a few times, with students reflecting on the feedback they received and educators reflecting on the data collected, coding and classifying it, both completing the "Reflective Observation" (RO) step.

The "Concept Development" and "Protocols' Adjustments" aligned the changes the students made to their projects based on the feedback they received and the experiments they conducted, and the educators adjusted the protocols for both facilitation and data collection as needed. Once the students completed the workshop, they continued working on their semester projects, as observed in the spiral turning into dotted arrows again at the outer layers. For the educators, results were collected based on students' project development, even after the workshop activities were completed, to ensure their entire concept development was followed through to the end of the semester. Including the related activities outside of the workshop timeframe allowed the research to consider the quality of the final projects and the degree to which the workshop activities have influenced decision-making.

The workshops' facilitated activities operated across different scales of engagement—from individual sensory provocation, asking students to reflect on their own design processes and the technology they used, to collective experimentation and the development of a group design strategy. Analog tools were introduced to the students through facilitated exercises to build confidence. The tools were always available for use, which further served as passive stimuli. Their presence invited students to engage voluntarily and intuitively, thereby promoting ownership over their design process and encouraging a culture of self-initiated learning.

Complementing the tools' passive role in the space, the structured workshop activities and assignments in both workshops were designed to introduce these tools in a facilitated manner, encouraging students to work with the technology and address any reservations they might have about using new tools. These sessions were not only instructional but also reflective, enabling students to recognize the potential of each tool. This was intentionally planned to encourage students to meaningfully embed their findings and reflections into their own design processes. The workshops were designed to promote independent exploration by gradually reducing the number of facilitated activities.

Group formation was another intentional strategy used to stimulate diverse cognitive engagement. Since the program itself is multidisciplinary, with a mix of architects and engineers, after following a mapping procedure to record the students' backgrounds, we were able to group them accordingly, ensuring an even distribution of competencies within each group. This enabled the observation of intra-group dynamics and the emergence of distinct modes of collaboration, co-creation, and knowledge exchange. By trying to understand how students from various backgrounds with different learning habits interact, the instructors were able to adapt the pedagogical approach to better meet the needs of individuals within a collective learning environment.

The two workshops used different designs to address various challenges. The main difference between the two workshops was the constraints on the tasks assigned to the students. In the first workshop, conducted in the Autumn semester of 2023, the students worked with what we called "open constraints," in which they were asked to shape a concept on a purely intuitive basis and test it through precise experiments by developing their own research questions. The first workshop lasted approximately 2 weeks, after which the students continued working on the same projects until the end of the semester. In this way, the workshop integrated with the course curriculum without disrupting regular semester activities or pivotal points, such as midterm presentations. Furthermore, because the final designs had no constraints other than incorporating well-developed climate-adaptive strategies, and because the research questions they had to answer were based on their own interests, the students continued using the tools after the workshop at their own discretion. We continued the observations past

the workshop timeframe and took the final projects as the culmination of the workshop.

The second workshop, conducted in the Autumn semester of 2024, was more intensive and had "closed constraints," where the students were asked to do the opposite of the previous semester, to run a precise experiment to get familiar with the climatic context, and then freely shape the concept on top of the results of the experiments. Intuition came after scientific correctness. The project they had to develop was still tied to the course curriculum, serving as a base for the semester project, but focused on optimizing a modular system, which was more of an exercise. The workshop took the project one step further than the previous year's by introducing a digital fabrication step and building a 1:1-scale model. Because of this, the time constraint for the first part of the workshop, dedicated to experimentation, was significantly shorter and more structured, with predefined research questions.

4.1.2 Moments of learning

Students were asked to apply their insights to the development of semester-long design proposals for climate-adaptive shelters for post-disaster recovery in the first workshop. During the second workshop, they were asked to optimize the Wiki House geometric system for designing a prefabricated, climate-adaptive unit across various climatic contexts, which served as an intensive two-week project preceding the students' semester-long project. Where the first project served as a platform for continuous testing, feedback, and refinement throughout the semester, making the transition between the workshop and the semester project relatively seamless, the second one aimed to allow students to reflect on their experience from the workshop and develop their own design process for their semester project.

Crucially, learning was not confined to moments of instruction or facilitation. It emerged organically during unstructured exploration, group interaction, and even social moments within the space. As students learned to recognize and articulate their own cognitive processes and to design strategies based on different tools, they, to some extent, developed both epistemic and practical understandings of the role of tools in their work. Considering our initial intentions with the workshops, which were to create and test a learning space that could foster a pedagogy successfully combining digital and analog tools for bioclimatic design, this conscious integration of different tools marks a hallmark of deep, transformative learning.

4.2 SPACE—the learning environment: a hybrid space for innovation

The physical space of the Climate HubLAB was intended to play a crucial role in reinforcing and facilitating the pedagogy we designed for the workshops, where experimental activities were combined with more conventional design workflows.

The lab (Figure 6) includes formal and informal learning environments, combining the functions of a traditional educational architecture studio, with desks where students can draw and discuss in groups their projects, and a research-oriented laboratory, where technical equipment makes it possible to build and test physical models based on experimental hands-on activities. The Lab serves moreover as a lecture room, with a side of the room equipped with an interactive

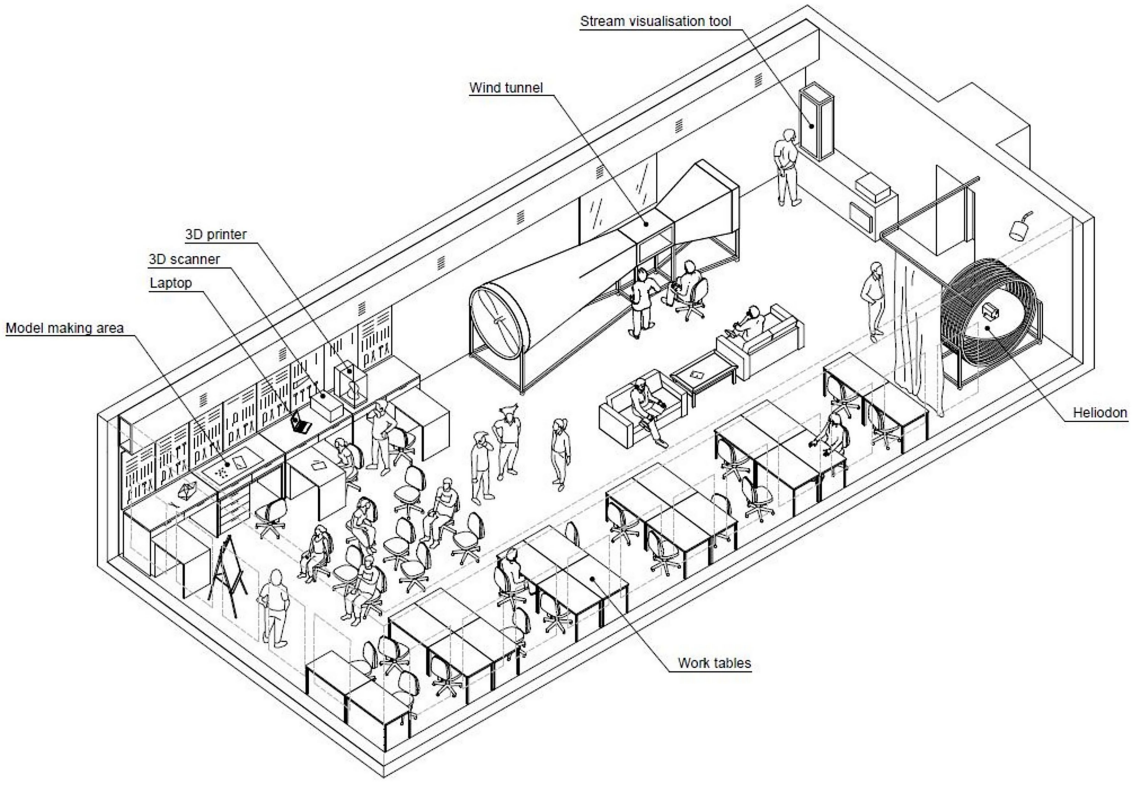


FIGURE 6
The Climate Hub-LAB conceptional drawing (picture by the MSc student Ilya Pugachenko).

projector, and as a presentation venue at the end of the semester, after the tables are rearranged. Social spaces further enriched the learning ecosystem. Kitchenettes, sofas, storage areas, bookshelves, and even a piano, brought in by the students, fostered a sense of community, relaxation, and shared ownership of the space. These elements actively supported informal peer learning and provided opportunities for interdisciplinary conversations and spontaneous collaboration.

Different zones in the studio were arranged to preserve each zone's identity while enabling students to transit between them easily. The conventional studio area, with desks and chairs, occupies the south wing of the HubLAB. Here, tables are arranged to facilitate students working in groups of four, with enough space for drawing, building conceptual paper models, and eventually hosting laptops on the side of hands-on activities. On the short side of the studio, towards the west, a foldable screen is coupled with an interactive projector. Here, students gather for lectures in the first part of the semester and common presentations, when projects are discussed in the second half. The wall to the north, without windows, offered space for model-making and testing, including high tables for hands-on model production and equipment to facilitate transfer of information between analog and digital environments, such as a 3D printer and a 3D scanner. Closer to the entrance, a manual heliodon allows testing smaller physical models under different sun patterns; a streamline visualization tool enables students to visualize air movement across conceptual models made of plexiglass. A gathering area near the entrance allows students to meet informally over coffee during a break, discuss projects, or simply share their learning experiences.

In a later development of the Climate HubLAB, the second studio at the disposal of the MSc in Sustainable Architecture was equipped with an automated heliodon for more advanced analyses about form and sun. This equipment required a larger, darker room and, therefore, the construction of a shell to create a dark space where more detailed models can be analyzed in relation to sun patterns and solar access. Different zones in the Climate HubLAB were intentionally kept separate but close enough for students to easily migrate between them. Students should, at first sight, understand which activities each zone can support and migrate from one zone to another "just in time" during the project development.

These spatial characteristics of the studio's design enabled an easy, almost seamless transition between structured learning, free experimentation, and unfacilitated discussions. This accessibility and intertwining of spatial functions were conceptualized to allow students to play and experiment at their own discretion, fostering spontaneous discovery. Individual desks, shared workstations, presentation setups, and analog simulation tools coexisted with a digital lab and software accessible for all students on their personal computers. This diversity of workspace typologies enabled students to choose the setting that best suited their activity, learning style, or design process phase. Furthermore, it enabled us to design a wide range of facilitated activities for the experimental workshops, allowing us to test students' engagement with individual tools or their responses to activities that combine multiple tools, both analog and digital.

The inclusion of analog simulation tools—strategically located throughout the space—provided necessary breaks from screen-based design work, both through facilitated and unfacilitated activities. The analog tools offered tactile, hands-on engagement with physical phenomena relevant to bioclimatic design adaptations, such as building orientation and levels of insolation for passive heating and cooling strategies, fenestration and shading strategies, wind analysis relevant to natural ventilation strategies, and spatial relationships. This analog-digital interplay not only diversified learning modalities but also grounded abstract digital representations accessible through the software used in the course in a tangible, embodied experience, which was our main objective throughout the study.

4.3 Technology—the climate HubLab: technology as an active medium

Technical equipment was positioned in the Climate HubLAB not merely as a computational tool for environmental analyses, but as a medium with agency. It served to help students connect disparate concepts, simulate individual climatic phenomena as opposed to the complex computational analysis that combines multiple climatic phenomena, and link theoretical frameworks to practical applications. Using various analog tools, including heliodons (both automated and non-automated), a water-stream visualization tool, and a wind tunnel, students could visualize and test their designs in relation to the sun and wind in real time. In addition to the space for conventional studio activities, the Climate HubLAB features stations for model-making equipment throughout the studio. The presence of 3D scanners and 3D printers aims to facilitate the exchange of information between the analog and digital environments throughout the design process. Activities employing CAD tools, such as Rhinoceros and parametric modeling tools (e.g., Grasshopper), in combination with Ladibug, Bumblebee, and Honeybee plugins, could then be included in a set of experiments, balancing hands-on activities and digital tools. The possibility to go back and forth between digital and analog tools created an iterative feedback mechanism aimed at supporting deeper modes of learning and more informed design decision-making. Technology thus became a bridge between abstract knowledge and real-world implications, allowing students to experience firsthand how design decisions impact the built environment and to test their choices in multiple ways.

Moreover, technology supported students in developing structured and informed approaches to design. It encouraged them to systematically investigate questions, validate assumptions, and integrate empirical findings into their projects. The hands-on activities with analog tools also opened the possibility for spontaneous, intuitive discovery and creative experimentation, aligning with Kolb's Learning Cycle (Kolb, 2021). This was all intentionally planned to facilitate a shift toward an intuition-driven learning dynamic, supported by the learning environment through evidence-based practice as opposed to the previous, 'software-driven' design.

4.4 User—the role of the student

Within the pedagogical structure established for the workshops and for the research conducted inside the DigiHands project, the students were not passive recipients of knowledge but active participants and co-constructors of it. Their role was imagined as that of an explorer, a self-actualizer, and a co-creator. In planning the two workshops, users were put at the center of teaching and learning activities to be tested. The two workshops were structured differently, and the results from the first encouraged us to tackle different challenges in the second.

In the first workshop, students were also encouraged to pursue their own lines of inquiry and pose their own research questions. As active learners, they were invited to take initiative and challenge assumptions. In the second workshop, the research questions were established from the outset, as the task required optimizing the same

geometric system across different climatic contexts. The workshop concluded with the prefabrication and construction of a prototype employing the analyzed construction system.

In both workshops, students were encouraged to reflect on their individual approaches to learning and to design critically, further reinforcing the shift from consumer to producer of knowledge, reorienting the traditional power dynamic in the studio, and empowering students to develop autonomy and agency.

5 Implementation and operation

Teaching and learning activities conducted in the Climate HubLAB were considered successful whenever they provided evidence of supporting students in achieving the course's learning outcomes. These include not only understanding the principles of climate adaptation but also developing the ability to integrate numerical concerns into creative form-finding processes. Feedback collected from students through questionnaires and interviews provided evidence that the technical equipment in the Climate HubLAB enabled free experimentation, often leading to unexpected results. Such practices engaged students in repeating experiments several times, thereby enhancing their understanding of the relationship between building design and climatic factors, and supported deep learning. As teachers and researchers observing activities in the laboratory, we observed that experiments were most successful in supporting learning when they aligned with the conceptual ideas underlying the project. To further support learning, experiments had to be simple enough to replicate in a limited amount of time; moreover, they had to be structured in a way that did not discourage critical thinking or limit students' creativity.

5.1 Pedagogy

Teaching and learning activities tested in the DigiHands workshops were structured following the four steps of the experiential learning cycle. Following Kolb's theories (Kolb, 2021), students were first asked to conduct a concrete experiment to solve a specific problem related to climate adaptation, then reflect on the experiment's results and conceptualize the outcome, and finally extract valuable principles to apply in the design project in a freer setting. During each experiment, following action research methods, we intervened whenever a learning opportunity appeared. At the end of each cycle, we discussed in plenum the results of the experiments and gave students honest feedback on the experiments and their translation into architectural concepts.

By employing the pedagogical methods we have designed for the workshops in the Climate HubLAB, and collecting and analyzing the data, it became apparent that the adopted pedagogical framework facilitated learning whenever students engaged in systematic design experiments driven by their own curiosity. Whether through the creation of intentional procedures to test a specific hypothesis or through open-ended exploration with analog tools, students experienced learning as an iterative, reflective process. Learning also manifested in translating these experiments into real-world projects.

A precise moment that illustrated the transition from supported practice to autonomy occurred during the midterm presentation of the first workshop. Initially, some students were cautious and hesitant as facilitators introduced the new analog tools. However, by the midterm presentation, most groups were freely experimenting with the manual heliodon with their group projects, pushing the boundaries of their initial concepts. They would structure new experiments and customize activities to support the development of their concepts. One example was a group project focusing on climatic adaptations in a hot, arid climate, in which students independently modified their initial models at an urban scale based on an unexpected finding they made with the manual heliodon. They conducted very detailed experiments with the machine, far beyond what was presented by the facilitators. Inspired by the results from these unguided experiments, which focused on orientation and positioning on an urban scale, the students in their final projects integrated unique solutions, such as successfully employing solar chimneys. This spike in autonomous engagement highlighted the successful fading of scaffolding, as students began to claim ownership of their learning and exploration.

5.2 Space

Experiences conducted in recent years allow for initial conclusions about the space and the formulation of informed hypotheses to be further investigated in the coming years. Overall, equipping the studio with machines and informal meeting areas contributed to creating a more inspiring atmosphere. Mimicking a conventional old-fashioned architectural design studio, where an indefinite number of models and sketches are produced throughout the design process, contributed to transforming the course into an experimental arena. Students aligned their behavior and activities with the studio, calling for model-making and testing. Experimental models were placed around and next to the machines for testing. Students perceived the studio as a maker-space where creativity comes into play, and numerous prototypes were being tested.

According to feedback collected in 2024 via Likert scales, 45% of the students "strongly agreed" and 30% "agreed" that the integrated studio was an inspiring arena that offered a constructive learning environment. Only 5% "disagreed" regarding that, while 20% remained "neutral." Most of the class agreed on the value of equipping the studio with machines for testing physical models, with 75% "strongly agreed," 20% "agreed," and 5% "neutral." Furthermore, 60% "strongly agreed" and 20% "agreed" that they benefited from the immediate and unlimited availability of the machines placed in the studio. 20% remained "neutral."

Parts of the studio are still unexploited today. Non-exploited areas were often used as storage for models produced throughout the semester. Over time, students took ownership of the studio, using available areas to create new informal spaces to meet, or even placing a piano or self-made wardrobes and furniture. Ownership was reflected in greater time spent in the studio and in engagement with collaborative design processes. In 2024, the activities in the Climate HubLAB were expanded to include the construction of a 1:1-scale prototype. TLAs were thus rephrased to move part of the activities into the model workshop at NTNU. There, students could produce prototypes and test assemblies before all the materials were moved and put together in the studio. Building in 1:1 scale was then moved into the studio, further increasing commitment, ownership of the space and time spent with peers in exploratory practices.

5.3 Technology

Experiences conducted in recent years have shown that user-friendly machines are most successful for integrating numerical concerns in free-form finding processes. More advanced equipment, however, stimulated curiosity and was successfully used whenever it was introduced in exercises that allowed students to build confidence first, before using it for their semester projects. During the last few years, by conducting the DigiHands workshops, we, as teachers and researchers, acquired valuable competencies that enabled us to adjust the timing and use of the equipment in a way that stimulated students to engage with it, take ownership of the process, and engage in deeper modes of learning. This was evident when evaluating the final semester projects, seeing how students conducted experiments on their own and integrated solutions based on the evidence they collected.

Employing a learning cycle loop for teachers and researchers (Figure 4, point 2) helped us adapt and refine the interventions and facilitated activities from the first to the second workshop. Putting open constraints on the workshop activities in 2023, leaving students to develop their own research questions and structure their own experiments, proved to be challenging when it came to the more complex simulation machines. The lack of confidence and the absence of structured, guided activities resulted in only two out of five groups using the automated heliodon. Learning from this experience, we restructured the experiments differently for the second workshop in 2024. Students were given instructions on how to build negative models made of plexiglass, on which they could place different shading masks, providing evidence of the optimal fenestration positions in the building. In the first year, we feared that starting the workshop with an analytical phase and explicit constraints could have limited students' creativity. However, testing a different approach, employing a structured analytical phase, helped students better frame the project's boundary conditions and build confidence in using the machines and in customizing experiments for different design purposes. In 2024, all students used advanced technical equipment more often than in 2023. Many students returned to use them several times after the workshop concluded to refine their models and project designs.

Feedback provided by students through Likert scales gave evidence of the importance of designing teaching and learning activities in a way that supports and benefits from the use of technical equipment placed in the Climate HubLAB. According to the Likert scales students completed in 2023 regarding the first module, using the conceptually clear heliodon, 75% found the module "very useful" or "useful," 10% found the activities informative, and the remaining 15% did not express an opinion. Furthermore, all five groups conducted experiments with the tool; they all claimed that they "made a discovery," that the activity made them more engaged through reflection, and that the use of the tool impacted their design choices. Regarding the second module, only two of five groups used the automated heliodon, and only one reported that the activity influenced their design choices. However, 50% of the students still found that learning how to use the tool was beneficial for future projects, 25% found the third cycle informative, 10% found it not helpful, and 15% did not express any opinion. In the last analyzed workshop, held in 2024, students were given better instructions on how to build physical models to use in the automated heliodon. Thanks to clearer guidelines, all students understood how to run experiments in this more advanced machine and managed to structure meaningful experiments that could support their design process. In 2024, 35% of the students "strongly agreed" and 25% "agreed" that the use of the heliodons made it possible for them to better grasp bioclimatic design principles. 10% "disagreed" and 30% remained "neutral."

Data collected during and after the workshop revealed that the extent to which students independently returned to use analog tools to run new, customized experiments depended not only on the complexity of the tools but also on how TLA were designed and facilitated. According to data recorded, 80% of students in 2023 and 70% in 2024 reported using the equipment to validate climate-adaptive strategies in their projects. Only a fourth of these, in 2023, used the automated heliodon again on their own initiative because it was considered too complex or time-demanding. TLAs were restructured in 2024, with more precise guidance on how quickly use the automated heliodon in more intuitive ways, producing conceptual inverted models in plexiglass. All students were trained, and 30% of the class used the machine again on an independent basis.

5.4 Users

The Climate HubLAB was initiated at the MSc program in Sustainable Architecture as a tool to empower students to run more informed design processes for climate adaptation. As a multidisciplinary program, the MSc in Sustainable Architecture is open to both architects and engineers. The program's teaching and learning activities are designed to facilitate collaboration. The Climate HubLAB plays a crucial role in this regard, enabling students to run experiments in which quantitative parameters are related to creative form-finding processes. Experiments facilitate collaboration, enhancing communication between engineers and architects by creatively translating numerical parameters into forms.

Students, as users of the Climate HubLAB, have always been at the center of the laboratory's development. The goal of merging conventional studios and research laboratories was to raise curiosity and desire for experimental design processes. As mentioned in the conception chapter, creating a space that could potentially enhance collaboration and communication between architects and engineers, The Lab enabled us to test pedagogical approaches that focus on fostering users' imagination, rather than developing content to be passively transferred to students. Using Kolb's Learning Cycle (Kolb, 2021) as a basis, we designed a learning experience that was structured enough to provide students with the basic knowledge to navigate the new tools they were introduced to, while also being flexible enough to allow students to take ownership of their design process. This created an iterative feedback loop that allowed us to adapt the activities according to the students' needs at any point of the workshops.

What is unique about the relationship between the user and the other three elements of the PST framework is that the user's agency can influence each of them, and in unexpected ways. This became evident throughout both workshops, as students were "empowered" by the maker space and transformed the studio to fit their own needs. Taking ownership and altering the studio with additional furniture and changing some of the organization, as mentioned earlier, is what we considered a success.

Through group discussions and presentations, students began to recognize how being intentional with their experiments shaped their approach to problem-solving and design iteration. This awareness was instrumental in enabling students to consciously structure their

design processes after the workshops and make informed decisions based on evidence through experimentation.

The students' testimonies at the end of the semester also indicated their appreciation for many aspects of the pedagogy employed. A challenging point for reflection was that, for some students, the lack of grading affected their level of engagement in the workshops, even though the target was the improvement and development of their semester project. However, those are personal attitudes towards the learning process, and only a few students had lower engagement because of them.

6 Results

Results of this work were traced, identifying discrepancies between the concept and the implementation of the Climate HubLAB, as described in chapters 4 and 5, respectively. Results are summarized in Table 4, where key elements related to Pedagogy, Space, and Technology are identified, supported by evidence collected in the workshops, and linked to challenges that may impact the development of teaching and learning activities within the space. In evaluating the capacity of the Climate HubLAB to support experimental pedagogies for bioclimatic design education, results from workshops conducted in 2023 and 2024 were considered, even though the last workshop's results relate more directly to pedagogy and technology, with slightly less emphasis on space. A summary of the results from both workshops is presented in Table 4.

A few examples from the students may illustrate their emerging learning trajectory.

Testimonials*: (Suggested intro to this testimonial: The following statements illustrate the value of access to the environment and the value of interaction with other students):

- "It was great to learn the tools outside of the semester project because we would not have taken the time later."
- "The experiments often sparked discussions which were very useful."
- "Honestly, if I were alone in the room with the model and the heliodon, I would not have found half as much; it would have been like sitting alone with the computer."
- "the workshop offered a deeper understanding of how to balance environmental responsiveness, technological innovation, and user-centered design in tackling climate challenges."

Testimonials**: (Again, the importance of access and the opportunity for informal exploration and experimentation is expressed):

- "We kept using the machines and used them much more than we thought we would. It was very convenient that they were just there."
- "The machines feel a lot more accessible because you do not need to build a complicated model and they are also just standing there."

Testimonials***: In these statements, the students demonstrate an understanding of the different affordances of digital and analogue tools and how their interplay enhances their understanding of "the bigger picture."

- "Even though the heliodon does not give you numerical values, it tells you something else."

TABLE 4 Summary of results.

Framework	Key elements promoting deeper levels of learning	Evidence	Challenges
Pedagogy	– Learner-centered Pedagogy, designed to follow the logic of Experiential Learning Cycles, designed to utilize specific technology and tools, and to inhabit a particular space	– The spontaneous repetition of the experiment and the ability to independently define experiments as a way to inform and support design choices were assumed as evidence of transformative learning – Testimonials*	– Structuring teaching and learning activities in a way that fits different learning styles – Motivating all students to participate in activities that are not graded
Space	– Space supporting diversity of learning styles by offering multipurpose spaces (Oblinger, 2005) – Merging formal and informal spaces supporting learning and discussions outside of the facilitated activities (Jamieson, 2005) – 24-h access promoting students' ownership of the space (Long and Ehrmann, 2005) – Optimized for particular activities with only relevant technology (Long and Ehrmann, 2005)	– Students perceived the studio as a maker-space and fitted their behavior and activities to it - making and testing physical models. – Students' initiatives to further furnish the studio gave evidence that students started to take ownership of the space and customize it in a way that fit and extended its use. – The space promoted learning by challenging students to become independent in their explorations and design, by conducting experiments on their own accord – Testimonials**	– unexploited areas contributing to a feeling of a not working space concept Part of the Lab equipment is still, to a large extent, unexploited today – Hard to trace conclusions based only on the data collected, because many unstructured observations provided more substantial evidence
Technology	– Technology serving as an enabler of free experimentation and spontaneous play – Incorporating analog simulation tools in hands-on activities, promoting analog breakouts as a contrast to the activities in the digital labs, which solely utilize software	– Students basing their design on discoveries made with the tools – Students conducting unfacilitated experiments and playing with the machines, testing solutions – Testimonials***	– Building solid experience in designing TLA such that students can grasp their potential before introducing them – Ensuring that digital tools employed in the course support the use of hands-on activities

– "The Heliodon made us see the bigger picture."
– "The analog simulation tools help you to understand things quicker and more intuitively than the software."

7 Discussion

Applying Radcliffe's Pedagogy–Space–Technology (PST) framework, and the extension developed by Manciaracina (PSTU) to the Climate HubLAB reveals strong interconnections among pedagogical intent, spatial design, and technological integration. We hope, with this study, to provide an example and inspiration for shaping future learning spaces for climate-adaptive design studios and to give directions for empirical validation of alternative ways to articulate learning spaces of a similar nature.

Pedagogically, the lab embodied a constructivist and experiential learning philosophy, with active student participation and reflective practice at its core. The pedagogy was dynamic and recursive, supported by a double-loop learning model and responsive action research methods.

The space itself was flexible, hybrid, and purposefully designed to support and invite a range of activities and learning styles. It blurred the boundaries between formal and informal learning, between individual reflection and group collaboration, and between digital and analog processes. Spatial design functioned not only as a container for learning but as a provocation—inviting action, experimentation, and social interaction.

Technology was meaningfully embedded as an enabler of connection, insight, and reflection. It was not treated as an end in itself but as a medium through which students could visualize, test, and understand complex systems. Its presence supported students in making research-informed decisions and strengthened the bridge between theory and practice.

It is challenging to discuss the results regarding only one factor - the space- because throughout this study, we have increasingly become aware of how interconnected all four aspects are, often overlapping across different categories, with equal importance and value. We cannot stress more how fundamental it is that all elements, Pedagogy, Space, Technology, and User, are taken into consideration and with equal priority when we attempt to curate learning experiences and try to learn from them. This is the main takeaway from this study.

8 Conclusion

The Climate HubLAB exemplifies a powerful approach to intentionally designed learning environments that foster self-directed, experiential, and reflective learning. By aligning pedagogy, space, and technology through a shared set of values and intentions, the lab successfully created a context where students could experiment freely, think critically, and take ownership of their learning journey. This case study affirms the value of designing educational environments that treat students as active agents, space as a pedagogical tool, and technology as a meaningful medium for exploration and knowledge construction. We hope this work can serve as an inspiration for further exploration of learning environments for design.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Written informed consent was not obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article because No names, ages, or other identifiable information were collected; all data are anonymous.

Author contributions

MS: Visualization, Formal analysis, Writing – original draft, Data curation, Investigation, Methodology, Conceptualization, Software, Writing – review & editing. LH: Writing – original draft, Formal analysis, Data curation, Writing – review & editing, Supervision, Conceptualization, Investigation, Methodology. LF: Methodology, Conceptualization, Validation, Investigation, Writing – review & editing, Supervision, Formal analysis, Visualization, Software, Project administration, Writing – original draft, Funding acquisition, Data curation, Resources.

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The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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